MASTERS OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA115

MANAGERIAL ECONOMICS

Unit - 8

Consumption Function and Investment Function

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1. INTRODUCTION

In the previous unit, we learnt about macroeconomics and some of itsmeasures. We also briefly examined the relation between income,consumption and savings. An increase in income leads to an increase in the level of consumption. However, the increase in consumption is normally not proportionate to the increase in the level of income. In this unit, we will study consumption function and investment function.

The consumption function explains the functional relationship that exists between income and the level of consumption. Psychological law of consumption, the average propensity to consume and the marginal propensity to consume help us to understand community behavior.

Investment function explains the relationship between aggregate income and aggregate investment. Various types of investment like; grossinvestment and net investment, public investment and private investment, autonomous investment and induced investment, marginal efficiency of capital and the rate of interest as the determinants of investment, give us aninsight into the nature of investment activity.

Multiplier is the ratio of change in income to a change in investment. Accelerator explains the increase in the level of investment as aconsequence of the increase in the level of income and consumption.

1.1 Objectives



After studying this unit, you should be able to:

- Describe the aggregative behavior of the economy as a whole.
- Explain the Keynes psychological law of consumption
- Evaluate the working of the multiplier and explain the importance of consumption in increasing the level of investment
- Apply the principle of acceleration, and judge the role of business expectations in determining marginal efficiency of capital and therefore investment.

2. CONSUMPTION FUNCTION

The consumption function indicates the relationship between consumption and income. Consumption is an increasing function of income.

According to Lord Keynes, the level of national output, income and employment directly depends on effective demand in an economy. The higher the level of effective demand, the higher would be the level of income and employment and vice versa. Effective demand consists of consumption expenditure and investment expenditure.

Consumption expenditure depends on the size of income and theconsumers' propensity to consume while investment expenditure depends on the marginal efficiency of capital and the rate of interest. Keynes suggested a high propensity to consume to tackle the problem of unemployment in an economy as one of the remedial measures. To understand the concept clearly it is necessary to distinguish between consumption and consumption function.

The term consumption refers to a particular amount of consumption out of a given amount of income. On the other hand, **consumptionfunction refers to different amounts of consumption at different levels of income.** It explains a functional relationship between changes in the levelof consumption as a result of changes in the levels of income. It indicates how consumption varies as income changes. If consumption is represented by C and income by Y then, the propensity to consume is C= f (Y). It implies that consumption is an increasing function of income. There is a direct relationship between the two. The higher the income, the higher would be the consumption and vice versa. Table 1 shows the relationship between income and consumption.

Table 1: Income versus Consumption

Income (Y) in Rs. crores	Consumption (C)
10	10
15	14
20	18
25	22
30	26

The table shows consumption as an increasing function of income and both the variables, Y and C, move in the same direction. Further, consumption is shown to change by Rs. 4 crores for each change of Rs. 5 crores in income.It is assumed that in the short run, the propensity to consume will remain stable. Increase in consumption is less than proportionate to the increase inincome.

When we represent this on a diagram, we get a curve rising upwards butless steeply. Figure 1 shows that the increase in consumption is smaller than the increase in income.

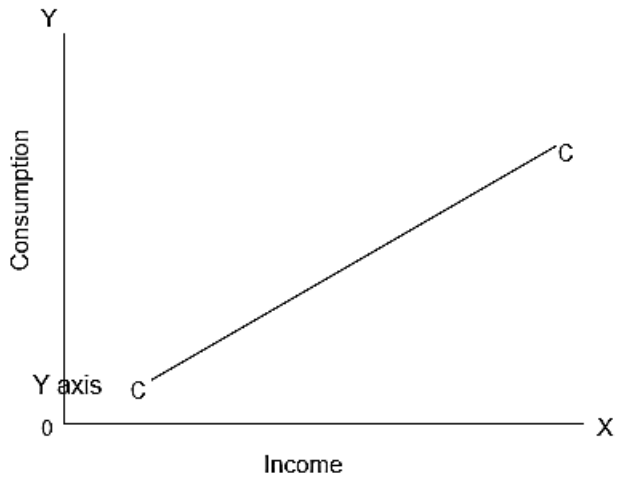


Figure 1: Graphical Representation of Income versus Consumption

In the diagram, the Y axis measures consumption and the X axis real income. The CC curve represents the consumption function.

Psychological law of consumption

In the words of Keynes "Men are disposed, as a rule and on the average, to increase their consumption as their income increases, but not by as muchas the increase in their incomes". In the short period, as the level of income of the people remains the same, the level of consumption also remains the same.

Generally, it is observed that when income increases, consumptionalso increases, but by a lesser proportion than the increase in income.

Suppose the total income of the community is Rs. 10 crore and the consumption expenditure are also Rs. 10 crores. In this case, there is no saving and investment. Further, the income increases to Rs.15 crore. Then, consumption also increases, but not to the extent of Rs. 15 crores. It may increase to Rs. 14 crore and Rs. 1 crore constitutes the savings. These savings creates a gap between income and consumption. This gap is in conformity with **Keynes psychological law of consumption, which**

statesthat, "when aggregate income increases, consumption expenditureshall also increase but by a somewhat smaller amount". This law tells us that people fail to spend on consumption, the full amount of increment in income. As income increases, the wants of the people get satisfied and they save more than what they spend. This law may be considered as a rough indication of the actual macro-behavior of consumers in the short- run. This is the fundamental principle of the Keynesian consumption function.

It is based upon his observations and conclusions derived from the study of consumption function. This law is also called the fundamental law of consumption. It consists of three interrelated propositions: -

- When the aggregate income increases, expenditure on consumption willalso increase but by a smaller amount.
- The increased income is distributed over both spending and saving.
- As income increases, consumption, spending and savings will go up.

As consumption expenditure progressively diminishes when income increases, a gap between income and expenditure arises. This tendency is so deep rooted in people's habits, customs, and the psychological set-upthat it is difficult to change in the short run. Hence, it is impossible to raise the propensity to consume of the people, so as to increase the national output, income and employment. Increasing the volume of investment in an economy can only fill the gap between income and consumption.

The average propensity to consume and the marginal propensity to consume

- ***The average propensity to consume (APC)*** **–** The relationship between income and consumption is measured by the average andmarginal propensity to consume. **The APC explains the relationship between total consumption and total income.** At a certain period of time, it indicates the ratio of aggregate consumption expenditure toaggregate income. **Thus, it is the ratio of consumption to income and is expressed as C/Y**.

  APC = TOTAL CONCUMPTION/TOTAL INCOME

  APC =C/Y = 8000/10000= .8 OR 80%

  Suppose the income of the community is Rs.10, 000 crore and consumption expenditure are Rs. 8,000 crores, then the APC is 8000/10,000 = 80% or 0.8. Thus, we can derive APC by dividing consumption expenditure by the total income.

- ***Marginal propensity to consume (MPC)* – MPC may be defined as the incremental change in consumption as a result of a given increment in income. It refers to the ratio of the change in aggregate consumption to the change in the level of aggregate income.** It may be derived by dividing an increment in consumption by an increment in income. Symbolically

  MPC = $\frac{\Delta C}{\Delta Y}$

  Suppose total income increases from Rs.10, 000 crore to Rs. 20,000 crore and total consumption increases from Rs. 8000 crores to Rs 15,000crore, then,

  MPC = $\frac{10,000}{7,000}$ = 0.7 or 70%

Technical characteristics of MPC

1. The value of MPC is always positive but less than one

   This means that when income increases, the whole income is not spent on consumption. Similarly, when income declines, consumption expenditure does not decline in the same proportion. Consumption expenditure never becomes zero.

2. MPC is greater than zero

   It is always positive. This means that an increase in income will lead to an increase in consumption. MPC cannot be negative.

3. MPC goes down as income increases.
4. MPC may rise, fall or remain constant, depending on many factors, both subjective and objective.
5. MPC of the poor is greater than that of the rich.
6. In the short-run, MPC is stable.

The concept of MPC throws light on the possible allocation of additional income between consumption and saving. It also tells us how the extra income will be divided in the Keynesian system. Saving, in the ultimate analysis, is equal to investment. In our example, MPC is 70% and as such, the MPS must be 30%. The reason is that the income of a community is divided between saving and spending. The sum of MPC and MPS equals 1.

Relationship between MPC and APC

Table 2 shows the relationship between APC and MPC.

Table 2: Relationship between APC and MPC

Rs. In crores

Income	Consumption	APC	MPC
100	100	100 %	–
200	180	90 %	80 %
300	240	80 %	60 %
400	280	70 %	40 %
500	300	60 %	20 %

1. Generally speaking, when income increases, APC as well as MPCdeclines but the decline in MPC is greater than APC.
2. As income increases, MPC falls and APC also falls but at a slower rate.
3. If MPC is rising, the APC will also be rising although at a slower rate.
4. When MPC is constant, APC may also remain constant.
5. APC, in some cases, may be equal to MPC. It is quite possible that 50% of the increased income is consumed and the remaining 50% is saved.
6. MPC is generally high in poor countries when compared to rich countries. In rich countries, the basic requirements are satisfied and therefore, the MPC would be less and MPS would be high. However, in poor countries, the majority of the people have to satisfy their basic needs and hence as income increases, the MPC also increases while MPS is generally low.

Factors determining consumption function

Broadly speaking, there are two factors which influence consumption function in the long run. They are:

- Subjective factors
- Objective factors

- ***Subjective factors*** **–** Subjective factors basically underlie and determine the form of the consumption. The subjective factors are either internal or endogenous in nature. **They mainly depend upon the personal decisions taken by the people.** Keynes has listed eight main

motives, which compel people to refrain from current spending. They are the motives of **precaution, foresight, calculation, improvement, independence, enterprise, pride and avarice.**

In addition to these factors, Keynes has also added a list of motives, which leads to consumption. "We could also draw up a corresponding list of motives to consumption such as **enjoyment, shortsightedness, generosity, miscalculation, ostentation and extravagance"**.

- ***Objective factors*** **–** Objective factors are those which depend on meritsand facts. In this case, personal factors will not come into picture. The following are some of the important objective factors which influence consumption:

1. Distribution of national income
2. Fiscal policy
3. Money income
4. Real income
5. Price and wage level
6. Changes in tastes and fashion
7. Changes in expectations
8. Wind fall (sudden) gains and losses
9. The level of consumer indebtedness
10. Attitude towards thrift
11. Liquid assets
12. Social and life insurance
13. Rate of interest
14. Business policies of corporations
15. Demonstration effect
16. Changes in expectations
17. Installment buying, etc.

The objective factors generally remain unchanged in the short period. Thus, propensity to consume in a short period is generally stable. It is becauseof this; Keynes relies on investment for the purpose of increasing employment during depression.

Importance of consumption function

It has got great theoretical as well as practical importance. Almost all countries of the world aim at removing unemployment, raising their national income and enjoying prosperity. For this purpose, a policy of planned economicdevelopment is essential. In the formulation of this policy, consumption function plays a very important role.

1. **It invalidates Say's law of markets**

   Say' law of markets which is the fundamental basis of classical theory of income and employment, states that" Supply creates its own demand". As aresult, there is no possibility of overproduction and unemployment. The consumption function tells us that the entire increase in additional income isnot spent on consumption goods. Hence, according to Keynes, supply,instead of creating its own demand, very often exceeds it and creates a glut of goods leading directly to overproduction and mass unemployment.

2. **It highlights the importance of investment in employment theory.**

   According to Keynes, in order to increase the volume of employment, bothconsumption and investment must be stepped up. However, consumptionfunction in the short run remains more or less constant and, as such, it maybe taken as given. Hence, investment plays a crucial role in determining thelevel of employment.

3. **It helps to explain the turning points of the business cycle**

   During the boom period, though income increases, consumption expenditure fails to match the increased incomes. Hence, saving increases-demand declines-and ultimately the slump develops. Similarly, during the period of slump, income contracts, people fail to reduce their expenditure on consumption to the full extent of the decrease in incomes. This tendency ultimately leads to a boom.

4. **It helps to explain the declining tendency of marginal efficiency of capital (MEC)**

   In rich advanced nations, the MPC is less than one. Hence, marginal efficiency of capital (MEC) shows a declining trend. This is because as income increases, expenditure falls, savings rise,

demand declines andproduction moves in the downward direction. Profits fell leading to a decline in MEC. Hence, investment declines and economic growth declines.

5. **It helps to explain the concept of secular stagnation**

Generally speaking, the MPC is low and MPS is high in most of the industrialized nations. The gap between income and consumption continuesto rise necessitating increase in investment. As propensity to consume is stable, propensity to save also tends to be stable. On the other hand, it becomes lesser with the lapse of time. The economy may sooner or later reach a stage where it finds itself unable to utilize fully and effectively its savings for the task of promoting full employment. Keynes calls such a situation as **"secular stagnation"**.

6. **It helps to find out the value of multiplier**

The value of multiplier is derived from the consumption function.

K = $\frac{1}{1-\text{MPC}}$ this helps us to understand the process of multiplication

Since the MPC is less than unity, the increase in national income is not in proportion to investment. The magnifying effect of multiplier declines due to a fall in consumption expenditure in an economy.

7. **It helps to explain the concept of underemployment equilibrium**

As MPC is less than one, the consumers fail to spend the full increased income on consumption. Effective demand becomes inadequate to bring about full employment equilibrium. Thus, the economy remains at underemployment equilibrium

8. **It upholds the state intervention**

Since overproduction and unemployment are possible owing to the deficiency of consumption, the economy cannot be a self-adjusting system by itself. Hence, state intervention is a must.

Thus, the consumption function helps us to analyze the process of income generation, expansion in employment opportunities, need for the state intervention and a very high level of investment to maintain the national income and employment.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	The relationship between consumption and income is called _____.
2	_____ is the ratio of change in total consumption to change in total income.
3	Two groups of factors that affect consumption function are _____ and _____.
True or False Questions:	
4	For the economy as a whole, consumption must equal saving. **(True/False)**
5	When a country saves a large portion of its income, it will have more capital and more productivity. **(True/False)**
6	Consumption, in economics, refers to household spending on durable and nondurable goods as well as household spending on services. **(True/False)**

3. INVESTMENT FUNCTION

Investment is the second important component of effective demand. InKeynesian economics, the term investment has a different meaning. In the ordinary language, it refers to financial investment. i.e. purchase of stocks, shares, debentures, bonds, etc. In this case, there is only transfer of rightsor titles from one person to another. It is an investment by one and disinvestment by another and as such, the value transaction mutually cancels out each other. They do not add anything to the total stock of capitalof the nation.

Investment, according to Keynes, refers to real investment. **It** implies the creation of new capital assets or additions to the existing stock of productiveassets. It refers to that part of the aggregate income, which is used for the creation of new structures, new capital equipment, machines, etc. that help in the production of final goods and services in an economy. Creation of income – earning assets is called investment. Thus, investment must generate income in the economy. Investment also refers to an addition to capital with such investment occurring when a new house is built or a new factory is built. Investment means making an addition to the stock of goods in existence. These activities necessitate the employment of more labor and thus result in an increase in national income and employment.

Investment is not a stock but a flow variable because it highlights the additions to the existing stock of capital. The productive ability of an economy is measured in terms of its stock of capital and its capacity to add to the existing stock of capital. Hence, it is a crucial factor in the economic development of any nation.

Types of investment

Keynes speaks of 5 types of investment. They are as follows:

1. **Private investment**

   It is made by private entrepreneurs on the purchase of different capital assets like machinery, plants, construction of houses and factories, offices, shops, etc. **It is influenced by Marginal Efficiency of Capital (MEC) and interest rates. It is profit – elastic.** Profit motive is the basis for private investment. Private entrepreneurs would take up only those projects which yield quick **results** and generally those that have a small gestation period.

2. **Public investment**

   It is undertaken by the public authorities like central, state and local authorities. It is made on building infrastructure of the economy, public utilities and on social goods, for example, expenditure on basic industries, defense industries, construction of multipurpose river valley projects, etc. In this case, the basic criterion and **motto is social net gain, social welfare and not profits.** The principle of maximum social advantage would govern public expenditure. It is also influenced by social and political considerations.

3. **Foreign investment**

   It consists of an excess of exports over the imports of a country. It depends on many factors such as propensity to export of a given country, foreigners' capacity to import, prices of exports and imports, state trading and other factors.

Induced investment

Figure 2 shows that as income increases, investment also increases and vice-versa.



Figure 2: Graphical Representation of Investment versus Income

Induced investment is another name for private investment. Investment, which varies with the changes in the level of national income, is called induced investment. When national income increases, the aggregate demand and level of consumption of the community also increases. In order to meet this increased demand, investment has to be stepped up in capital goods sector which finally leads to increase in the production ofconsumption goods Therefore, we can say that **induced investment is income – elastic** i.e., it increases as income increases and vice-versa.

Thus, it is sensitive to changes in income and is **governed by profit – motive.** The shape of the induced investment curve has been shown as rising upwards to the right. This means that as income increases, investment also increases and vice-versa.

4. Autonomous investment

Autonomous investment is another name for public investment. **The investment, which is independent of the level of income, is called as autonomous investment.** Such investments do not vary with the level of income. Therefore, it is called income-inelastic. It does not depend on changes in the level of income, consumption, rate of interest or expected profit. Figure 3 depicts that though income changes, investment more or less remains constant.

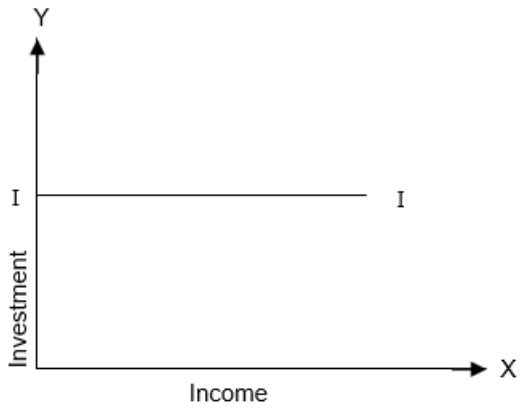


Figure 3: Autonomous Investment Curve

Autonomous investment depends upon population growth, technologicalprogress, discovery of new resources, etc. For example, expenditure on public buildings, transport and communications, defense, public utilities,water supply, generation of electricity, etc are considered as autonomous investment. It is guided by social welfare rather than profit motive.

The autonomous investment curve is perfectly inelastic. And as such it indicates that though income changes, investment more or less remains constant.

There are a few other concepts of investment. They are as follows:

- ***Gross investment*** **–** Gross investment refers to the total real investmentor addition to capital stock of the country.
- ***Replacement investment*** **–** A part of gross investment that is used for replacing the old capital equipment is called replacement investment.

- ***Net investment*** **–** The net investment is equal to the gross investment minus replacement investment. Hence, net investment = gross investment – capital consumption or replacement investment.
- ***Ex-ante investment*** **–** The investment that is intended or expected or planned is known as ex-ante investment.
- ***Ex-post investment*** **–** The actual or realized investment is known as ex-post investment.

Determinants of investment

Investment decisions taken by entrepreneurs depend upon a number of factors like interest rate, level of uncertainty, political environment, rate of growth of population, level of existing stock of capital, and the necessity of new products. It also depends on investors' level of income, level of inventions and innovations, level of consumer demand, availability of capital and liquid assets of the investors, government policy, etc.

It is necessary to note that investment is more volatile and unpredictable. Itis highly unstable in the short run because the factors determining it are highly complex and uncertain in their nature. The above-mentioned factors no doubt generally affect the volume of investment. However, the most important inducement to invest is the consideration of the profit. The profitability of investment depends mainly on two factors:

- Marginal efficiency of capital (MEC)
- Interest rate (IR).

It relates to the cost-benefit analysis. The businessman while investingcapital has to calculate the cost of borrowing and the expected rate of profitsfrom it.

4. MARGINAL EFFICIENCY OF CAPITAL AND BUSINESS EXPECTATIONS

Marginal efficiency of capital

It refers to productivity of capital. It may be defined as the highest rate of return over cost accruing from an additional unit of capital asset. It also refers to the yield expected from a new unit of capital. The MEC in its turn depends on two important factors:

- Prospective yield from the capital asset and
- Supply price of capital assets.

The MEC is the ratio of these two factors. The prospective yield of a capital asset means the total net return expected from the asset over its lifetime. After deducting the variable costs like cost of raw materials, wages, etc from the marginal revenue productivity of capital, an investor can estimate the prospective income (expected annual returns and not the actual returns) from the capital asset. Along with it, the investor also has to consider the supply price or replacement cost of the capital asset.

The supply price of a capital asset is the cost of producing a brand-new asset of that kind, not the supply price of an existing asset. It is the actual amount of money spent by an investor while purchasing new machinery or erecting a new factory.

The MEC of a particular type of asset means what an investor expects to earn from an additional unit of it compared with what it costs him. To be more specific, MEC is the rate of discount, which will make the presentvalue of the capital assets equal to their future value (prospective yield) in their lifetime i.e. Supply price = discounted prospective yield.

The MEC can be calculated with the help of the following formula:

$$Cr = \frac{Q1}{(1+r)^1} \quad \frac{Q2}{(1+r)^2} \quad \frac{Q3}{(1+r)^3} \quad \frac{Qn}{(1+r)\,n}$$

In the above formula Cr represents the supply price or replacement cost of the new capital asset. Q1, Q2, Q3 indicate the prospective yields in the various years 1 2 3 ... n and r represent the rate of discount which will make the present value of the series of annual returns just equal to the supply price ofcapital asset. Thus, r denotes the rate of discount or MEC.

We can illustrate the meaning of MEC as a rate of discount by means of a simple arithmetical example. Let's assume that the supply price of a capital asset is Rs.3000/- and the asset will become useless after two years. Furthermore, the capital assets is expected to yield Rs.1100/- at the end of one year and Rs.2420/- at the end of 2 years. Now, it is obvious that the rate of discount of 10% will equate the future yields of the asset with its current supply price. At a 10% discount rate, the present value of Rs.1100/- discounted for one year plus Rs.2420/- discounted for 2 years amounts to an aggregate sum of Rs.3000/- which is the supply price of the capital asset. The above-mentioned formula can be used to explain the same point

$$Cr = \frac{Q1}{(1+r)^1} + \frac{Q2}{(1+r)^2} \quad 3000 = \frac{1100}{(1+0.1)^1} \quad \frac{2420}{(1+0.1)^2}$$

$$3000 = \frac{\text{Rs. } 1100}{[1.10]} + \frac{\text{Rs. } 2420}{[1.10]^2} = \frac{1100}{1.1} + \frac{2420}{1.21}$$

$$3000 = \text{Rs.}1000 + \text{Rs.}2000$$

In this case, the discounted prospective yield is equal to the current supply price of the capital asset. If the expected rate of yield is greater than the supply price, it becomes profitable to invest and otherwise not.

The volume of induced investment depends on MEC and IR. It is necessary to note that:

- **When MEC > IR, the effect on investment is favorable**
- **When MEC < IR, the effect on investment is adverse**
- **When MEC = IR, the effect on investment is neutral**

Generally speaking, the MEC of a capital fall as investment increases and the reasons for this are as follows:

1. The prospective yields of the asset will fall as more and more units of it are produced. This happens because as more assets are produced,they will compete with each other to meet the demand for the product and consequently, their general earnings will decline.
2. The operation of the law of diminishing marginal returns.
3. Higher investments create higher demand for capital assets leading toan increase in the supply price of capital assets. Consequently, the total production cost rises. Thus, MEC declines with

an increase ininvestment either as a result of decreasing prospective yield or increasing supply price of capital asset.

4. Higher investment results in higher production, a reduction in per unit cost,lower price for the products and lower earnings from the sales.

Thus, the MEC falls as investment increases because costs go up and earnings fall. The fall in MEC will be different at different levels of investment. The MEC curve drops downwards from left to right and this tendency can be explained with the help of the following example. Table 2 depicts an example of changes in MEC in relation to IR and Investment.

Table 3: Changes in MEC of capital in relation to IR and Investment

The IR in % p.a	**Volume of investment in crores**	**MEC of capital in % p.a**
13%	5000	13%
11%	7000	11%
9%	9000	9%
7%	11000	7%
5%	13000	5%
3%	15000	3%

On the OX axis, we represent different amounts of investment and on OY axis, we represent MEC and IR. The ME curve indicates the MEC. It can be seen that as investment increases, the ME curve slope downward. It is clearthat if the current IR is 9 %, then the entrepreneurs will invest Rs. 9000 crores because at this point the MEC is also 9% i.e. MEC = IR. If the IR fallsto 7%, then the entrepreneurs will invest Rs 11000/-. This is because the MEC is also 7% at this point. Figure 4 shows Investor's return

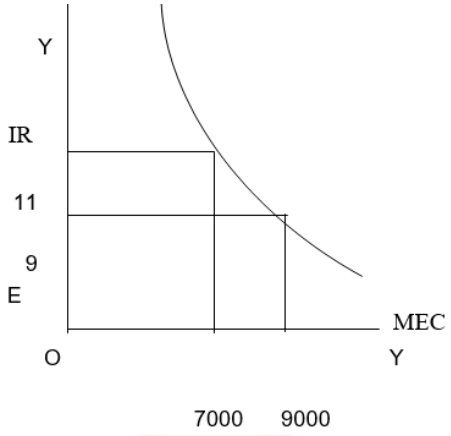


Figure 4: Investment

The MEC represents an investor's return and the IR is the cost. Obviously, the return on capital must be equal to its cost. Thus, the MEC and IR are closely related to each other and they move together. We can conclude that given a MEC curve, the investment will depend on the existing IR in the market.

Determinants of MEC

Several factors that affect MEC are as follows:

I. Short run factors

1. **With** the expectation of increased demand, higher MEC leads to largerinvestment and vice versa.
2. **Cost and price:** If the production costs are expected to decline and market prices to go up in future, MEC will be high leading to a rise in investment and vice versa.
3. **Higher propensity to consume,** leading to a rise in MEC encourages higher investment.
4. **Changes in income:** An increase in income will stimulate investment and MEC while a decline in the level of income will discourage investment.
5. **Current state of expectations:** If the current rates of returns are high, the MEC is bound to be high for new projects of investment and vice- versa. This is because the future expectations to a very great extent depend on the current rate of earnings.
6. **State of business confidence:**

   During the period of optimism (boom), the MEC will be generally high and during period of pessimism (depression), it will be generally less.

II. Long run factors

1. **Rate of growth of population:** In a capitalist economy, a high rate of population growth leads to an increase in MEC because it leads to an increase in the demand for both consumption and investment goods. Onthe contrary, a decline in the population growth depresses MEC.
2. **Development of new areas:** Development activities in the new fields like transport and communications, generation of electricity, constructionof irrigation projects, ports, etc would lead to a rise in MEC.
3. **Technological progress:** Technological progress would lead to thedevelopment and use of highly sophisticated and latest machines, equipment and instruments. This will add to the productive capacity of the economy leading to an increase in MEC.
4. **Productive capacity of existing capital equipment:** Under utilised existing capital assets may be fully utilized if the demand for goods increases in the economy. In that case, the MEC of the same asset will definitely rise.
5. **The rate of current investment:** If the current rate of investment is already high, there would be little scope for further investment and as such the MEC declines.

Thus, several factors both in the short run and in the long run affect theMEC of a capital asset.

Hence, several measures are to be taken to stimulate private investment in an economy.

In the Keynesian theory, investment is a very important and strategic variable. In order to increase the volume of national output, income and to tackle the problem of unemployment, the remedial measure suggested by Lord Keynes is to raise the level of investment in an economy. It is seen thatas the income of the community increases, consumption also increases but by less than the increase in income. Hence, in order to have sufficient demand to sustain an increase in employment, there must be an increase in real investment equal to the gap between income and consumption out of income. In other words, employment cannot increase unless investment increases.

Role of business expectations in determining MEC

Business expectations play a vital role in determining MEC and therefore investment. Level of income and employment in an economy are determined by two factors: propensity to consume and inducement to invest.Of the two, propensity to consume is more or less stable. Fluctuations in income and employment, therefore, depend mainly on the inducement to invest. The inducement to invest in turn depends on the rate of interest and the marginal efficiency of the capital. Since the rate of interest

is relatively stable or sticky, fluctuations in investment depend primarily upon the changes in the MEC. There are two determinants of the MEC: the cost of the capital asset and, the rate of return from the asset.

Uncertainty in the prospective yield or business expectations causes instability in MEC. As business expectations change, the volume of investment changes and this causes changes in business activity and employment.

Expectations regarding the prospective yield of capital assets are of two types: (a) Short - term expectations (b) Long - term expectations.

Short-term expectations are based on the existing stock of capital and the intensity of consumers' demand for the goods, which are known and remain more or less stable.

On the other hand, long-term expectations relate to future changes in the size of the stock of capital assets and changes in the level of aggregate demand which are uncertain. Thus, the long-term expectations are highly unstable, but are more important in explaining fluctuations in investment and employment.

The long-term expectations are influenced by the following factors:

1. **The state of confidence –** How certain and confident are businessmen with regard to the future change.
2. **Stock exchange valuation –** The value attached to it by the dealers in the stock exchange.
3. **Irrevocable decisions –** Decisions made by bold and dynamic entrepreneurs.
4. **Elements of instability –** Frequent changes in the assessment of the prospects of various investments have introduced a lot of changes in investment activity.
5. **Link with investments –** Stock exchange dealings influence new investments by establishing links between the new investments and the present investments.
6. **Behavior of investors –** Since there is mass valuation of assets on the stock exchange, there are alternating waves of pessimism and optimism.

Apart from these, political events like war, elections, etc. also influence the prospective yield of the capital assets.

Thus, investment decisions are made in an uncertain atmosphere, based onbusiness expectations with regard to the marginal efficiency of capital.

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
7	Investment made by government and departmental undertakings is called _____ investment.
8	Depreciation is part of _____.
9	Depreciation is part of _____.
10	_____ varies with the changes in the level of national income.
True or False Questions:	
11	In macroeconomics, investment refers to the purchase of mutual funds,stocks, or bonds. **(True/False)**
12	If the government encourages savings, it would lead to lower interestrates and higher investments. **(True/False)**
13	When the government has a deficit, it leads to lower investment andhigher interest rates. **(True/False)**

5. MULTIPLIER

Meaning and working of multiplier

Prof. Kahn developed the concept of **"Multiplier"** with reference to employment. Lord Keynes, on the lines of employment multiplier, developedan **"Investment Multiplier".** It is derived from the concept of marginalpropensity to consume, and refers to the effects of changes in investment outlays on aggregate income through consumption expenditure. It has acquired greater significance in recent years to explain the process of income generation in an economy when the volume of investment changes.

There are various types of multipliers such as; income multiplier, investment multiplier, employment multiplier, foreign trade multiplier, etc.

Keynes' investment multiplier is based on the fundamental psychological law of consumption. It states that as income increases, consumption also increases but less than proportionately. Consequently, the consumptiondemand in the short run remains constant and it cannot be increased. The alternative to increasing the output is to increase the volume of investment. Even though consumption function is a major determinant of aggregate demand, it is not the prime initiator of changes in income and output. The changes in the volume of investment will bring about changes in the level of income and output. How changes in income are affected can be understood with the help of investment multiplier.

Multiplier may be defined as a ratio of change in income to a change in investment. It expresses the relationship between an initial increment in investment and the final increment in income. It shows by how many times, the effect of an initial change in investment is multiplied by causing changes in consumption and finally in the aggregate income. Change in investment generally gives rise to change in income by a multiple amount. Whenever anadditional investment is made in the economy, it increases the aggregate income not only by an amount equal to the additional investment but by somewhat greater than that. The logic is simple. The original investment increases income not only in the industry where investment is made, butalso in certain other industries whose products are demanded by people employed in investment goods industries. For example, if an increase in investment of Rs. 5 lakhs cause an increase in income of Rs. 25 lakhs,then the multiplier would be 5. If the increase in income is Rs. 30 lakhs, thenthe multiplier would be 6. Algebraically, this relationship can be expressedas follows:

$$\Delta K = \frac{\text{Change in Income}}{\text{Change in Investment}} = K = \frac{25}{5} = 5$$

where delta (Δ) stands for change or increase, K for multiplier, Y for income and I for investment respectively.

The size of the multiplier is directly derived from the size of the MPC. The higherthe MPC, the higher would be the size of the multiplier and vice versa. The multiplier is equal to the reciprocal of 1 minus MPC. The formula to calculatethe size of the multiplier is as follows:

$$K = \frac{1}{1-MPC}$$

If we know the size of the MPC, the value of the multiplier can be easilyfound out. If MPC is 2 / 3, then multiplier would be as follows:

$$K = \frac{1}{1-MPC} = \frac{1}{1-2/3} = \frac{1}{1-1/3} = 3$$

We know that MPC + MPS =1. If we deduct MPC from 1 we get MPS.Hence, the above equation can be expressed in the following manner:

$$K = \frac{1}{MPS}$$

If the MPC is 9 / 10, deducting 9 / 10 from 1, we get 1 / 10. This is the MPS. The reciprocal of 1 / 10 is 10, which is the value of the multiplier. In short, the multiplier is the reciprocal of the MPS, which is always equal to 1 minus the MPC.

From the above explanation, it is clear that:

- The higher the value of MPC, the higher would be the value of K and vice-versa.
- When MPS = 0 and MPC =1, then there will be a 100 percent increasein income every time or the multiplier effect will be continuous.
- When MPS =1 and MPC = 0, then what is earned will be saved and thevalue of multiplier will be equal to 0.

Working of the multiplier and the process of income generation

The process of income generation through the working of the multiplier canbe expressed in the following manner:

Assumptions

- MPC = ½ or K = 2.
- An investment of Rs 10 crores will generate an income of Rs 20 crores.

Table 4 shows Impact of Multiplier on Incomes.

Table 4: Impact of Multiplier on Incomes

Period or rounds	**Investment in crores (Rs)**	**Change in income incrores (Rs)**
1	10	10
2	10	05
3	10	2.50
4	10	1.25
5	10	0.62
6	10	0.31
	Total	20

The multiplier process is based on the principle that one man's expenditureis another man's income. If MPC of one individual is high, the income of another man is also high. From the respective table it is clear that as we move from one round to another, the initial investment gives rise to adwindling series of successive increments in income because MPC is generally less than one.

Assumptions

- MPC = 2 / 3 or MPS = 1 / 3 or K = 3.
- Original investment is Rs 30 crores.
- Additional investment is Rs 10 crores.

As the multiplier is 3, the additional investment of Rs.10 crores leads to an increase in the income of the community to Rs.30 crores. This can be understood with the help of the following diagram.

In the respective diagram, QR represents original investment of Rs 30crores. SS is the saving curve, which intersects the investment curve at the point M, which indicates the original income of the community at Rs 130 Crores. Q1R1 is the new investment line, which indicates an additional

investment of Rs 10 crores. The new investment curve Q1R1 intersects the same saving curve at M1. At this new equilibrium point, the income of the community is Rs 160 crores. It is clear that as a result of an additional investment of Rs 10 crores, income has gone up by 3 times (from Rs 130 crores to Rs 160 crores). Figure 5 shows Results of investments

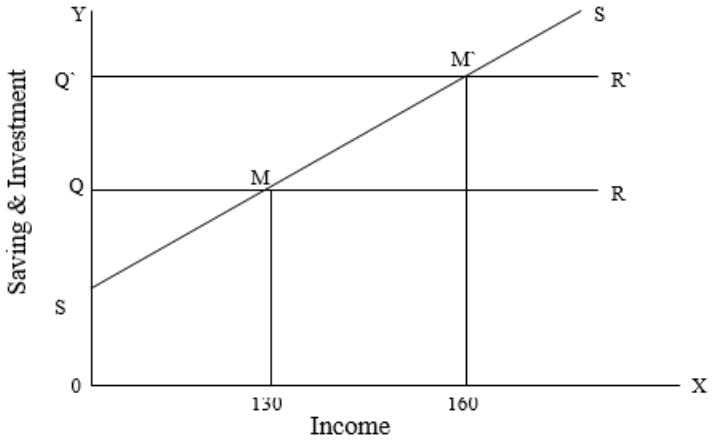


Figure 5: Impact of multiplier on Savings and Investments

Assumptions and limitations of the multiplier

1. **Availability of consumer goods**

   Multiplier works satisfactorily if the volume of goods and services on which the additional income may be spent are available in plenty. Otherwise, people are unable to spend their income on them. Consequently, MPC falls leading to a decline in the value of K.

2. **Maintenance of investment**

   In order to realize the full value of K, it is necessary that the various increments in investment be repeated at regular intervals. In case, it is not done, it will not be possible to raise the income to the multiplier level.

3. **Net Increase in investment**

   In order to get the full value of K, there should be a net increase in investment. Increase in investment in one sector of the economy should not be neutralized by a decrease in investment in another sector of the economy. Otherwise, the working of the multiplier is obstructed.

4. **No change in the size of MPC**

   In the process of income generation, there should not be any change in the value of MPC. If there is any change in the size of MPC, then the value of K also changes.

5. **No investment from induced consumption**

   In the multiplier theory we analyze only the impact of investment on consumption, but the reverse, accelerator is totally ignored. If the accelerator is allowed to operate and effects of induced consumption on investment are also taken into account, then the value of the multiplier would be far greater and would also be achieved at an earlier stage in the process of income generation.

6. **No time gap between successive expenditure on consumption**

   If there is a gap between receipt of income and expenditure even in the short run, the full value of the K cannot be realized because as MPC falls, the size of the K also declines.

7. **Existence of a closed economy**

   If there is trade between different countries, the value of K may be restricted by the amount of excess of imports over exports. A part of the total money will go out of the country if there are imports and to that extent, the value of the K declines.

8. **Existence of less than full employment condition**

   If the economy is working at full employment level, there is no scope for an increase in output, income and employment even though investment increases.

9. **It is based on the number of assumptions**

   These assumptions may not be found in practice. Consequently, the 'actual' multiplier may be greatly restricted and will be different from the 'ideal' multiplier.

10. **No direct relation between investment and income**

    There is no precise, pre-determinable or mechanical relationship between social income, consumption, investment and extent of employment.

11. **Keynes multiplier is a static concept**

    It shows the process of income propagation from one point of equilibrium to another and that too under static conditions. It gives little insight into the actual process by which the economy achieves a new equilibrium.

Prerequisite conditions for the working of the multiplier

1. **Existence of involuntary unemployment**

   National output, income and expenditure can be increased by additional investment only when there is involuntary unemployment condition in an economy. Otherwise, there will be no scope for expansion in employment even if investment increases.

2. **Existence of an industrial economy**

   Multipliers can freely operate in an industrial economy rather than in an agricultural economy because the demand for industrial goods is relatively stable than that of agricultural goods and as such, the MPC and the value of K will be high.

3. **Existence of excess capacity in consumer goods industries**

   A high level of investment will succeed in utilizing the unutilized and underutilized excess capacity to the extent possible. This leads to the creation of more employment.

4. **Existence of elastic supply of other factor inputs in the market**

   A higher investment would result in higher output, income and employment only when the other supplemental factor inputs are available in abundance.

Leakages in the multiplier

Income not spent on consumption is called **'leakage'** in the cumulative income stream. This leakage obstructs the increase in output and income. These leakages will arise on account of the following reasons:

- ***Savings*** **–** Higher the level of savings, the lower would be the value of Kand vice-versa.
- ***Accumulation of idle cash balances*** **–** If people keep more idle cash balances with them, then the MPC declines and the value of K declines.
- ***Debt cancellations*** **–** If people use a part of their income to repay their old debts, then the current MPC declines and the value of K also declines.
- ***Purchase of old shares and stocks*** **–** A part of the income may be spent on buying old stocks, shares and other securities in the market. This would lead to a decline in the current MPC and a decline in the value of K also.

- ***Imports*** **–** Payments on imports would reduce domestic consumption leading to a decline in the value of K.
- ***Price inflation*** **–** Inflation would reduce the purchasing power of the people and consequently the MPC and also the value of K.
- ***Taxes*** **–** Higher taxes would reduce the income of the people, MPC and also the value of K.
- ***Corporate savings*** **–** Undistributed profits of joint stock companieswould reduce the incomes of the shareholders, their MPC and thus the value of K.

If all these leakages are properly plugged in the income stream, the effect ofmultiplier in the process of income generation would be higher, taking the economy towards full employment level.

Practical importance of the multiplier

1. It is a major tool of macroeconomic theory focusing attention on investment as the significant element in increasing the level of employment.
2. It summarizes the working of the entire Keynesian model.
3. It describes how income is generated in an economic system like stonecausing ripples in a lake.
4. It is a valuable guide to public investment policy.
5. It is helpful for framing a suitable full employment policy.
6. It has great practical significance in formulating an anti-cyclical policy to smoothen business fluctuations in an economy. Thus, it is necessaryfor the study of the trade cycle, its trends and control.
7. According to Prof. Samuelson, the multiplier theory explains why an easy money policy is ineffective and deficit spending is effective during the period of depression.
8. For increasing the income and employment, investment should bestarted in a sector where the multiplier may be greater.
9. It is used for explaining expansion in different fields of economic activities, for example; credit multiplier, budget multiplier, etc.
10. It upholds the government intervention, active participation and macroeconomic management relating to income, output, employment, etc.\

11. It helps to understand how equality between saving and investment is brought about. An increase in investment leads to an increase in income. Consequently, saving also increases and becomes equal to investment.
12. It emphasizes the significance of deficit financing. Increase in public expenditure by creating deficit budget helps, in creating income and employment by multiple times the initial increase in expenditure.

Thus, the concept of multiplier has not only brought about a revolution in economic theory but also in framing various economic policies. Keynes regards it as a path-breaking contribution to economic theory.

6. ACCELERATOR

The principle of **"accelerator or acceleration"** is another important tool of economic analysis. It is older than multiplier. The accelerator dates back to 1914 or even before. It is associated with the name of Prof. J. M. Clark, an American economist who was mainly responsible for popularizing it in 1917.

Multiplier and accelerator are two parallel concepts. The multiplier concept is inadequate to explain the process of income generation in a complete manner. It shows the effect of investment only on consumption expenditure and how the increase in investment will bring about an increase in national income through multiplier effect. In short, multiplier explains the effects of investment on consumption and how the volume of consumption depends on the volume of investment. The multiplier fails to analyze the effect of increase in consumption on investment. In order to know how consumption affects the volume of investment, we have to study the concept of accelerator.

Accelerator shows the effect of changes in consumption on induced investment and tells us how the volume of investment depends on the level of consumption. The combined action of both multiplier and accelerator will clearly explain how the aggregate national income increases as a result of the increase in the volume of investment in an economy. When the incomes of the people increase, purchasing power and the demand for consumption goods increase. In order to produce more consumer goods, more capital goods are required. This leads to an increase in the demand for investment in capital goods industries. Thus, a rise in income leads to a rise in induced investment in capital goods industries. For example, if an expenditure of Rs.4 crores on consumption goods industries, leads to an investment of Rs. 12 crores in capital goods industries, then, we can say that the accelerator is 3. Production of consumer goods requires a particular amountof capital goods. Therefore, the accelerator is generally more than one. In order to produce consumer goods sometimes, more capital is not neededand sometimes, it is not required at all because the existing stock of capital goods becomes sufficient. Hence, accelerator may be less than one or zero. The accelerator is called **"Magnification of derived demand"** because investment depends on employment.

In order to understand the effect of accelerator, it is necessary to know the acceleration co-efficient. The ratio between the net change in consumption expenditure and the induced investment is called 'Acceleration Co-efficient". Symbolically,

a = Δ I / Δ C where,

a stand for acceleration co-efficient

Δ I = net change in investment expenditure

Δ C = net change in consumption expenditure

The working of the accelerator can be explained with the following imaginaryexample. Let us assume that in order to produce 1000 units of consumer goods, 100 machines are required. The capital output ratio in this case is 1:10. Further assume that the working life of a machine is 10 years and after 10 years the machine has to be replaced. It implies that every year, 10 machines have to be replaced in order to maintain the constant flow of 1000 units of consumer goods. Hence, the acceleration coefficient in this case is 1 and the annual demand for machines will be 10. This is called **"Replacement Demand".**

Now let us assume that the demand for consumer goods goes up by 10%. Consequently, more machines are required now to meet the increased demand for consumer goods. Now we require 10% or 10 new machines. (10% of 100 machines are 10. Hence, the total demand for machines will now be 20. It means a 100% increase in the demand for machines (10 for replacement and another 10 for meeting the increased demand). The investment in the capital goods industry has doubled, because in our example, the value of the accelerator is 10.

Acceleration effects investment

Table 5 depicts Total investments

Table 5: Impact of accelerator on induced and total investments

Period	Consumption goods	Capital goods required	Investment needed for replacement	Induced investment	Total investment	% Change in total investment
0	1000	100	10	Nil	10	–
1	1000	110	10	10	20	100

From the table it is clear that a 10% increase in demand for consumption goods has resulted in a 100% increase in total investment outlay, as accelerator is one. Figure 6 shows the working of the accelerator.

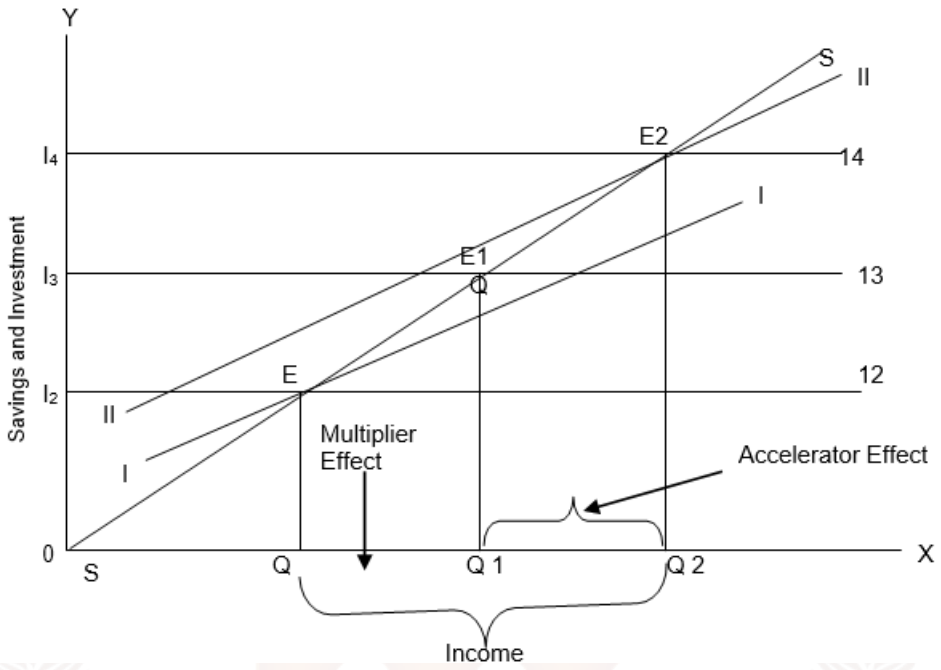


Figure 6: Effect of accelerator in investments

In the diagram, SS and II represent saving and investment curves. E indicates the original equilibrium position where S and I meet each other. OQ is the original equilibrium income.

Now, investment increases from I2 to I4. Consequently, the national income increases from OQ to OQ2. The jump in income is Q to Q2. If the increase in investment from I2 to I4 had been purely public investment, then the entire increase in income Q to Q2 would have been due to only the multiplier effect. However, the Q to Q1 increase in national income is due to the multiplier effect because the increase in investment from I2 to I3 is public investment.

The increase in national income from Q1 to Q2 is due to the acceleration effect because the increase in investment from I3 to I4 is due to induced investment. The total multiplier and accelerator effect on income is measured by QQ2 (QQ1 due to multiplier effect and Q1 Q2 due to acceleration effect).

Limitations of accelerator

1. There should be no excess capacity in capital goods industries. If there is excess capacity, additional production of consumer goods does not require additional capital goods.
2. If there is an excessive number of machines, there is no need for further investment in capital goods industries.

3. If the demand for consumption goods is purely temporary in nature, producers will not make any additional investment in capital goods industries. On the other hand, they make use of existing machines more intensively.
4. In many cases, investments made in capital goods industries do not await changes or increase in consumption, for example, investment in public sector industries.
5. If adequate financial resources are not forthcoming to capital goods industries, in spite of the increase in consumption, production of capitalgoods cannot be increased.
6. It is always wrong on our part to expect a constant ratio between production of consumer goods and capital goods.

   In all these cases, the value of the accelerator may not be fully realized.

Practical importance

1. It explains the process of income generation more clearly as it takes intoaccount the effect of consumption on investment.
2. It explains why fluctuations in income and employment occur ratherviolently.
3. It tells us why capital goods industries fluctuate much more than consumption goods industries.
4. It helps in understanding the different phases of business cycles very clearly. However, accelerator, by itself, cannot completely explain the entire reason for trade cycles.

SELF-ASSESSMENT QUESTIONS – 3

Fill in the blanks:	
14	The ratio of change in national income resulting from a change inautonomous investment is called _____.
15	When MPC = 1, multiplier will be _____.
16	shows the effects of consumption on investment; whereas_____ shows the effect of investment on consumption.
17	Accelerator is called _____
True or False Questions:	
18	As the cash reserve ratio increases, the money multiplier increases.**(True/False)**

19	The amount of money in the economy partly depends on the behaviorof banks. **(True/False)**
20	If the reserve ratio is 20%, Rs. 500 can be created from Rs. 100 ofreserves. **(True/False)**

Activity - 1

Collect the information on investments made by any company and study its various factors like the rate of interest, marginal efficiency of capital, level of uncertainty, political environment, rate of growth of population, level of existing stock of capital, inventions, consumers' demand, etc.

7. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- Macroeconomic concepts of consumption function, investment function, multiplier, accelerator, etc. explain vividly the functioning of an economy.
- Consumption function refers to the schedule of propensity to consume atvarious levels of income.
- The psychological law of consumption states that consumption tends to rise with income, but less proportionately.
- Average propensity to consume is the ratio of consumption to income and is expressed as C/Y. Marginal propensity to consume is the ratio of the change in consumption to the change in the level of income and is expressed as Δ C/ Δ Y.
- Consumption function is determined by a number of subjective andobjective factors. The concept explains the obstacles in the attainmentof full employment equilibrium. It explains the turning points of the business cycles, declining tendency of MEC, secular stagnation, underemployment equilibrium and upholds the importance of state intervention and increased investment in the generation of employment and income. The value of the multiplier is derived from the consumption function.
- Investment refers to real investment, denoting an addition to real capital assets as well as to the wealth of society. There are various kinds of investment like private investment, public investment, foreign, induced, autonomous, gross, net, etc.
- Investment is determined by a number of factors like the rate of interest, marginal efficiency of capital, level of uncertainty, political environment, rate of growth of population, level of existing stock of capital, inventions, consumers' demand, etc. Investment is highly unstable in the short run. Inducement to invest mainly depends on the rate of interest and the marginal efficiency of capital.
- The MEC depends on (a) the prospective yields i.e., expected profitability of the capital assets, and (b) the replacement cost of these assets. Business expectations play a very important role in determining MEC and therefore investment.

- Multiplier is the ratio of change in income to a given change in investment. There are a number of limitations to the working of the multiplier and it presupposes the existence of involuntary unemployment, industrial economy, excess capacity in consumer goods industry, etc.
- A number of leakages like savings, imports, taxes, etc. obstruct the increase in output and income. It describes how income is generated in an economic system like stone causing ripples in a lake. It is a valuable guide to public investment policy.
- Accelerator explains the effect of increase in consumption to the demand for capital goods and the related investment. Acceleration depends on the capital output ratio and the durability of the capitalassets. It explains the process of income generation more clearly as it takes into account the effect of consumption on investment.

8. GLOSSARY

Accelerator	-	Changes in induced investment due to change in consumption.
Autonomous investment	-	The investment which is independent of the level of income.
Induced investment	-	Investment, which varies with the changes in the level of national income.
Marginal efficiency of capital	-	The highest rate of return over cost accruing from an additional unit of capital asset.
Marginal propensity to consume	-	The incremental change in consumption as a result of a given increment in income.
Multiplier	-	Ratio of change in income to a change in investment.

9. TERMINAL QUESTIONS

1. What is the consumption function?
2. State the importance of consumption function.
3. What is investment function; discuss the various factors that determinethe investment function.
4. Discuss the working of the multiplier and explain the various leakages inmultiplier.
5. Discuss the concept of accelerator and explain its working and what itsuses are.

10. ANSWERS

Self-Assessment Questions

1. Consumption function
2. Marginal propensity to consume
3. Subjective and objective factors
4. False
5. True
6. True
7. Public
8. Gross investment
9. Autonomous
10. Induced investment
11. False
12. True
13. True
14. Multiplier
15. Infinite
16. Multiplier; accelerator
17. Magnification of derived demand
18. False
19. True
20. True

Terminal Questions Answers

Answer 1: Consumption function refers to different amounts of consumption at different levels of income. It explains a functional relationship between changes in the level of consumption as a result of changes in the levels of income. Refer to section 2.

Answer 2: Importance of consumption function is that it invalidates Say's law of markets, highlights the importance of investment in employment theory, explains the turning points of the business cycle, the declining tendency of marginal efficiency of capital (MEC), the concept of secular stagnation, upholds the state intervention, finds out the value of multiplier etc. Refer to section 2.

Answer 3: Investment, according to Keynes, refers to real investment. It implies the creation of new capital assets or additions to the existing stock of productive assets. It refers to that part of the aggregate income, which is used for the creation of new structures, new capital equipment, machines, etc. that help in the production of final goods and services in an economy. Investment decisions taken by entrepreneurs depend upon a number of factors like interest rate, level of uncertainty, political environment, rate of growth of population, level of existing stock of capital, and the necessity of new products. It also depends on investors' level of income, level of inventions and innovations, level of consumer demand, availability of capital and liquid assets of the investors, government policy, etc. Refer 3.

Answer 4: Multiplier is the ratio of change in income to a given change in investment. A number of leakages like savings, imports, taxes, etc. obstruct the increase in output and income. It describes how income is generated in an economic system like stone causing ripples in a lake. It is a valuable guide to public investment policy. Refer section 4.

Answer 5: Accelerator explains the effect of increase in consumption to the demand for capital goods and the related investment. Acceleration depends on the capital output ratio and the durability of the capital assets. It explains the process of income generation more clearly as it takes into account the effect of consumption on investment. Refer section 5.

11. CASE STUDY

Looking Ahead

Alok Ray

A low and stable inflation may reduce uncertainty about the future cost and profitability of investment projects in India.

Despite all the hype about returning to a 9 per cent growth path, all important policy makers have now got resigned to the fact that the GDP growth rate of India would, at best, be in the neighbourhood of 7 per cent in the fiscal year 2011-12 and 7.5-8 per cent in 2012-13.

Even this is contingent in the global economic situation not worsening further. Both international and domestic factors have contributed to this step-down. First, the global slowdown (especially in EU, USA and Japan) has adversely hurt our exports, despite some diversification of our export markets away from the western world in recent years.

Second, the anti-inflationary monetary policy has hurt growth by raising the cost of loan-financed investment and consumption expenditure. Third, the global uncertainty along with the policy paralysis in India, (thanks to the various scams and the consequent unwillingness of officials to clear projects and the coalition politics standing in the way of taking policy decisions like; allowing FDI in retail and hiking prices of diesel and power) has made investors – both domestic and foreign – to a 'wait and see' attitude towards investing in India.

Inflation (WPI), staying at around 10 per cent for most of the last year,has moderated to some extent, mainly because of a very substantial decline in food inflation (which at one time reached 20 per cent) in recent weeks. Though the current negative food inflation may not last long as this is primarily due to seasonal factors, the government officials are hoping that the headline (WPI) inflation would stabilise to below 7 percent by March 2012. If this materialises, the RBI may ease its monetary policy which would help growth. A low and stable inflation may also reduce some of the uncertainty about, the future cost and profitability of investment projects in India, thereby encouraging more capital investment.

Our balance of payments is under severe stress due to steady rise in the international price of petroleum products, slowing of export growth andnet outflow of portfolio funds.

The rupee fell by some 14 per cent against the dollar in 2011, making the rupee the worst performing currency in the emerging world. The depreciation of the rupee has further contributed to inflation (cost-push)by raising the rupee cost of imported crude and raw materials. It also increased the government's oil and fertiliser subsidy bill, worsening the already bad fiscal deficit situation. The food security bill to be introduced soon will further add to the fiscal woes. The depressed stock market doesnot leave room for any substantial disinvestment proceeds to finance the fiscal deficit. Additional government borrowing from the market (about Rs 50,000 crores) to finance the unanticipated rise in deficit would exert an upward pressure on interest rates.

Major Hurdles

Some of the big banks are in trouble, after lending money to loss making airlines which have eventually become bad loans. In addition, PSU bankshave financed 70 per cent of accumulated Rs. 82,000 crore losses of power distribution companies. Power companies are not generating power to capacity; either because they do not have money to pay for coalor the coal companies are unable to supply the required quantity of coal. The mining output has gone down after many scams involving illegal mining have come to light. Big power and mining projects are stalled due to problems in land acquisition and environmental concerns. Overall, the infrastructure – especially power and water – remains the major hurdle in realising our growth potential.

Are these temporary issues? Well, one can always point to the investment ratio of 36 per cent and savings ratio of 34 per cent (along with a capital output ratio of 4:1) to argue that India can potentiallyachieve a 9 per cent growth on a sustained basis, even without the helpof foreign capital. India has the further advantage of having an abundant supply of skilled manpower at relatively low cost and the demographic advantage of a higher ratio of young people in the population.

China, after growing at double digit for more than two decades, may now slow down due to rising labour cost (as their supply of cheap labour is being exhausted) and an appreciating yuan as China, is gradually forced (by western political pressures and domestic pressures to give a better standard of living to Chinese workers) to switch from export-led to domestic consumption-led growth. All these may open up new opportunities for India to fill up some of the vacuum left by China, especially in the global market for low-cost manufacturing goods.

Burgeoning subsidy bill is crowding out much needed public investment in irrigation, roads, rural electrification, education, health and extension services in agriculture which would have helped both growth and inclusion in a sustained manner.

*(**Source:** Based on an article of the same title that appeared in the Deccan Herald on Jan 19, 2012)*

Discussion Questions:

1. What is the relation between inflation and profitability of projects?
2. How has the rupee's depreciation impacted interest rates?
3. How have subsidies influenced investments in India?

Hint: Use the theoretical concept and answer the questions

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